Федеральное государственное бюджетное образовательное учреждение
высшего образования
«Чувашский государственный университет им. И.Н. Ульянова»
Кафедра вычислительной техники

Лабораторная работа № 3

По дисциплине “Защита информации”

Выполнил: студент группы ИВТ-42-23

Иванов И.Л.

Проверил: кандидат технических наук

Ковалев С.В.

Чебоксары, 2026

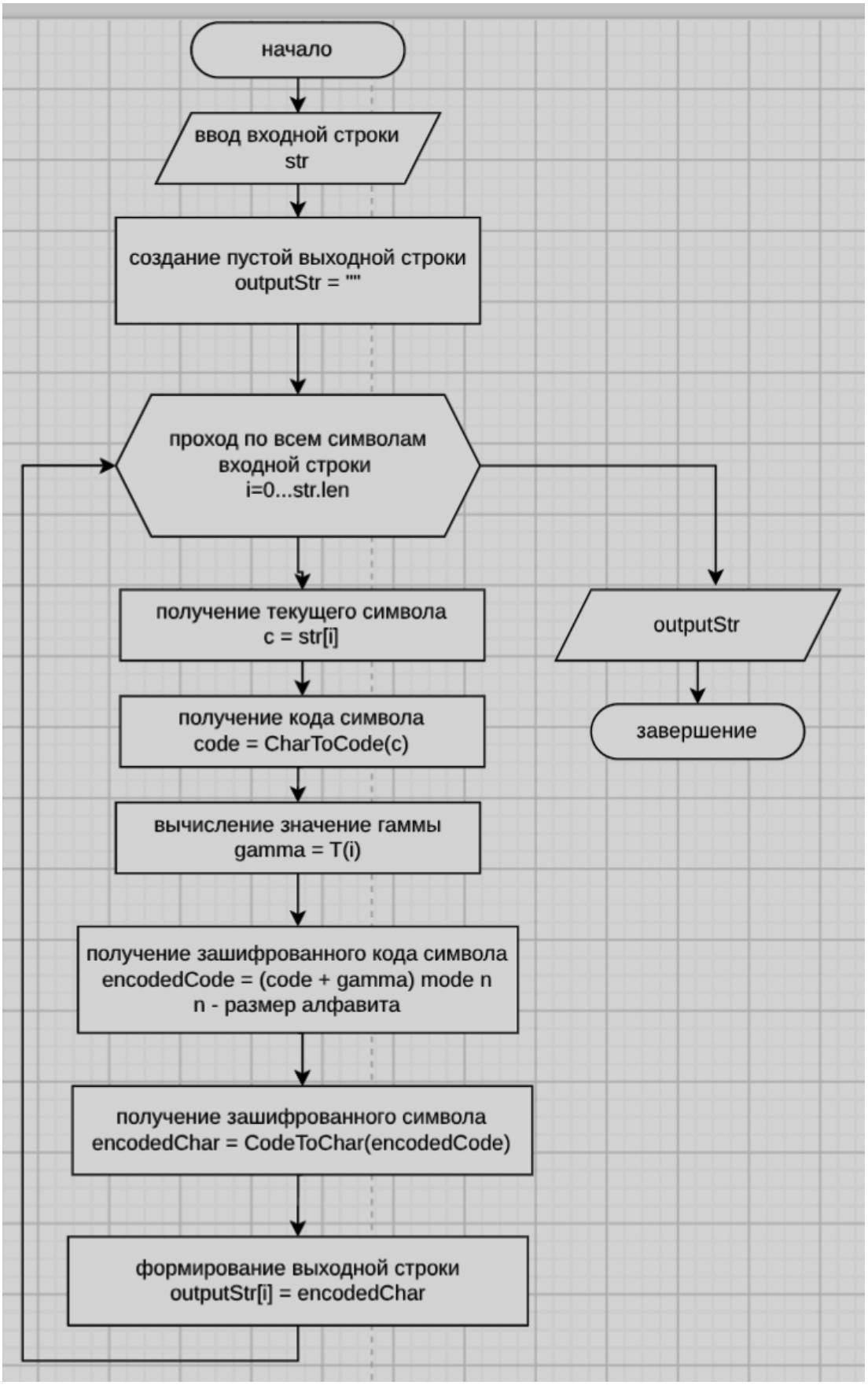
1. **Цель работы:**

знакомство с классическим криптографическим алгоритмом - алгоритмом линейного шифрования данных (шифрования гаммированием).

2. **Индивидуальное задание.**

Задание			
№ п.п.	*A*	*C*	*T(0)*
1	5	51	13

3. **Текст программы, реализующей индивидуальное задание.**



https://github.com/kukirmash/labs/tree/main/7sem/ZI/lab3

```
#include "utils.h"

#include <time.h>

#define ALPHABET_SIZE 256
#define TESTS_COUNT 10
#define MAX_LEN 50

#define B ALPHABET_SIZE
#define A 5
#define C 51
#define T0 13

//-------------------------------------------------------------------------
-----------------------
// T(i+1)=(A*T(i)+C) mod B,
int T(u_int8_t i)
{
   if (i == 0)
       return T0;
   else
       return (A * T(i - 1) + C) % B;
}

//-------------------------------------------------------------------------
-----------------------
u_int8_t CharToCode(char c)
{
   return (u_int8_t)c;
}

//-------------------------------------------------------------------------
-----------------------
unsigned char CodeToChar(u_int8_t code)
{
   return (unsigned char)code;
}

//-------------------------------------------------------------------------
-----------------------
std::string EncodeString(const std::string& inputStr)
{
```

```
   std::string outputStr(inputStr.size(), '\0');

   for (int i = 0; i < inputStr.size(); i++)
   {
       unsigned char c = inputStr[i];
       u_int8_t code = CharToCode(c);

       u_int8_t gamma = T(i);

       u_int8_t encodedCode = (code + gamma) % ALPHABET_SIZE;
       unsigned char encodedChar = CodeToChar(encodedCode);

       outputStr[i] = encodedChar;

        printf("Char: %c | Code: %3d | EncodedCode: %3d | EncodedChar:
%c\n", c, code, encodedCode, encodedChar);
   }

   return outputStr;
}

//-------------------------------------------------------------------------
-----------------------
std::string DecodeString(const std::string& encodedStr)
{
   std::string outputStr(encodedStr.size(), '\0');

   for (int i = 0; i < encodedStr.size(); i++)
   {
       unsigned char c = encodedStr[i];
       u_int8_t code = CharToCode(c);

       u_int8_t gamma = T(i);

       int temp = code - gamma;
            u_int8_t decodedCode = (temp >= 0) ? temp : (temp +
ALPHABET_SIZE);
       unsigned char decodedChar = CodeToChar(decodedCode);

       outputStr[i] = decodedChar;

        printf("EncodedChar: %c | EncodedCode: %3d | DecodedCode: %3d |
DecodedChar: %c\n", c, code, decodedCode, decodedChar);
```

```
    }

    return outputStr;
}

//------------------------------------------------------------------------
----------------------
int main()
{
    setlocale(LC_ALL, "");
    srand(time(0));

    int strLen = 0;
    std::string inputStr = "";

    strLen = rand() % MAX_LEN + 1;

    inputStr = GenerateRandomCharString(strLen);

    printf("\n");
    printf(inputStr.c_str());
    printf("\n");
    std::string encodedStr = EncodeString(inputStr);
    printf("\n");
    std::string decodedStr = DecodeString(encodedStr);
    printf(decodedStr.c_str());

    return 0;
}

//------------------------------------------------------------------------
----------------------
```

4. **Пример открытого текста и соответствующей ему шифрограммы.**

ȝxX�ZC�3�F}?Bk��8�an ���9�!|vT"��a�=9�u�

Char: ȝ | Code: 198 | EncodedCode: 211 | EncodedChar: Ó

Char: x | Code: 120 | EncodedCode: 239 | EncodedChar: ï

Char: X | Code: 88 | EncodedCode: 222 | EncodedChar: Þ

Char: ◆ | Code: 190 | EncodedCode: 143 | EncodedChar: ◆

Char: Z | Code: 90 | EncodedCode: 162 | EncodedChar: ◆

Char: C | Code: 67 | EncodedCode: 222 | EncodedChar: ◆

Char: ◆ | Code: 245 | EncodedCode: 47 | EncodedChar: /

Char: 3 | Code: 51 | EncodedCode: 136 | EncodedChar: ◆

Char: ◆ | Code: 165 | EncodedCode: 129 | EncodedChar: ◆

Char: F | Code: 70 | EncodedCode: 197 | EncodedChar: ◆

Char: } | Code: 125 | EncodedCode: 43 | EncodedChar: +

Char: ? | Code: 63 | EncodedCode: 216 | EncodedChar: ◆

Char: B | Code: 66 | EncodedCode: 114 | EncodedChar: r

Char: k | Code: 107 | EncodedCode: 142 | EncodedChar: ◆

Char: ◆ | Code: 216 | EncodedCode: 186 | EncodedChar: ◆

Char: ◆ | Code: 239 | EncodedCode: 140 | EncodedChar: ◆

Char: 8 | Code: 56 | EncodedCode: 124 | EncodedChar: |

Char: ◆ | Code: 179 | EncodedCode: 58 | EncodedChar: :

Char: a | Code: 97 | EncodedCode: 55 | EncodedChar: 7

Char: Code: 27 | EncodedCode: 124 | EncodedChar: |

Char: ◆ | Code: 156 | EncodedCode: 180 | EncodedChar: ◆

Char: n | Code: 110 | EncodedCode: 25 | EncodedChar:

Char: | Code: 16 | EncodedCode: 154 | EncodedChar: ◆

Char: ◆ | Code: 228 | EncodedCode: 201 | EncodedChar: ◆

Char: ◆ | Code: 173 | EncodedCode: 89 | EncodedChar: Y

Char: ◆ | Code: 160 | EncodedCode: 47 | EncodedChar: /

Char: | Code: 127 | EncodedCode: 125 | EncodedChar: }

Char: Code: 27 | EncodedCode: 68 | EncodedChar: D

Char: | Code: 8 | EncodedCode: 8 | EncodedChar:

Char: O | Code: 79 | EncodedCode: 130 | EncodedChar: ◆

Char: ◆ | Code: 225 | EncodedCode: 19 | EncodedChar:

Char: ◆ | Code: 194 | EncodedCode: 239 | EncodedChar: ◆

Char: ? | Code: 199 | EncodedCode: 219 | EncodedChar: ?

Char: 9 | Code: 57 | EncodedCode: 208 | EncodedChar: ?

Char: ? | Code: 128 | EncodedCode: 166 | EncodedChar: ?

Char: ! | Code: 33 | EncodedCode: 18 | EncodedChar:

Char: | | Code: 124 | EncodedCode: 100 | EncodedChar: d

Char: v | Code: 118 | EncodedCode: 49 | EncodedChar: 1

Char: T | Code: 84 | EncodedCode: 46 | EncodedChar: .

Char: " | Code: 34 | EncodedCode: 151 | EncodedChar: ?

Char: ? | Code: 188 | EncodedCode: 56 | EncodedChar: 8

Char: ? | Code: 209 | EncodedCode: 112 | EncodedChar: p

Char: a | Code: 97 | EncodedCode: 175 | EncodedChar: ?

Char: ? | Code: 255 | EncodedCode: 184 | EncodedChar: ?

Char: = | Code: 61 | EncodedCode: 13 | EncodedChar:

Char: 9 | Code: 57 | EncodedCode: 124 | EncodedChar: |

Char: ? | Code: 238 | EncodedCode: 112 | EncodedChar: p

Char: u | Code: 117 | EncodedCode: 50 | EncodedChar: 2

Char: ? | Code: 236 | EncodedCode: 208 | EncodedChar: ?

EncodedChar: ? | EncodedCode: 211 | DecodedCode: 198 | DecodedChar: ?

EncodedChar: . | EncodedCode: 46 | DecodedCode: 186 | DecodedChar: ?

EncodedChar: ? | EncodedCode: 239 | DecodedCode: 120 | DecodedChar: x

EncodedChar: ? | EncodedCode: 222 | DecodedCode: 88 | DecodedChar: X

EncodedChar: ? | EncodedCode: 143 | DecodedCode: 190 | DecodedChar: ?

EncodedChar: ? | EncodedCode: 162 | DecodedCode: 90 | DecodedChar: Z

EncodedChar: ? | EncodedCode: 222 | DecodedCode: 67 | DecodedChar: C

EncodedChar: / | EncodedCode: 47 | DecodedCode: 245 | DecodedChar: ?

EncodedChar: ? | EncodedCode: 136 | DecodedCode: 51 | DecodedCode: 3

EncodedChar: ? | EncodedCode: 129 | DecodedCode: 165 | DecodedChar: ?

EncodedChar: ? | EncodedCode: 197 | DecodedCode: 70 | DecodedChar: F

EncodedChar: + | EncodedCode: 43 | DecodedCode: 125 | DecodedChar: }
EncodedChar: ◆ | EncodedCode: 216 | DecodedCode: 63 | DecodedChar: ?
EncodedChar: r | EncodedCode: 114 | DecodedCode: 66 | DecodedChar: B
EncodedChar: ◆ | EncodedCode: 142 | DecodedCode: 107 | DecodedChar: k
EncodedChar: ◆ | EncodedCode: 186 | DecodedCode: 216 | DecodedChar: ◆
EncodedChar: ◆ | EncodedCode: 140 | DecodedCode: 239 | DecodedChar: ◆
EncodedChar: | | EncodedCode: 124 | DecodedCode: 56 | DecodedChar: 8
EncodedChar: : | EncodedCode: 58 | DecodedCode: 179 | DecodedChar: ◆
EncodedChar: 7 | EncodedCode: 55 | DecodedCode: 97 | DecodedChar: a
EncodedChar: | | EncodedCode: 124 | DecodedCode: 27 | DecodedChar:

ncodedChar: ◆ | EncodedCode: 180 | DecodedCode: 156 | DecodedChar: ◆
EncodedChar: | EncodedCode: 25 | DecodedCode: 110 | DecodedChar: n
EncodedChar: ◆ | EncodedCode: 154 | DecodedCode: 16 | DecodedChar:
EncodedChar: ◆ | EncodedCode: 201 | DecodedCode: 228 | DecodedChar: ◆
EncodedChar: Y | EncodedCode: 89 | DecodedCode: 173 | DecodedChar: ◆
EncodedChar: / | EncodedCode: 47 | DecodedCode: 160 | DecodedChar: ◆
EncodedChar: } | EncodedCode: 125 | DecodedCode: 127 | DecodedChar:
EncodedChar: D | EncodedCode: 68 | DecodedCode: 27 | DecodedChar:

ncodedChar: | EncodedCode: 8 | DecodedCode: 8 | DecodedChar:
EncodedChar: ◆ | EncodedCode: 130 | DecodedCode: 79 | DecodedChar: O
EncodedChar: | EncodedCode: 19 | DecodedCode: 225 | DecodedChar: ◆
EncodedChar: ◆ | EncodedCode: 239 | DecodedCode: 194 | DecodedChar: ◆
EncodedChar: ◆ | EncodedCode: 219 | DecodedCode: 199 | DecodedChar: ◆
EncodedChar: ◆ | EncodedCode: 208 | DecodedCode: 57 | DecodedChar: 9
EncodedChar: ◆ | EncodedCode: 166 | DecodedCode: 128 | DecodedChar: ◆
EncodedChar: | EncodedCode: 18 | DecodedCode: 33 | DecodedChar: !
EncodedChar: d | EncodedCode: 100 | DecodedCode: 124 | DecodedChar: |

EncodedChar: 1 | EncodedCode: 49 | DecodedCode: 118 | DecodedChar: v
EncodedChar: . | EncodedCode: 46 | DecodedCode: 84 | DecodedChar: T
EncodedChar:  | EncodedCode: 151 | DecodedCode: 34 | DecodedChar: "
EncodedChar: 8 | EncodedCode: 56 | DecodedCode: 188 | DecodedChar: 
EncodedChar: p | EncodedCode: 112 | DecodedCode: 209 | DecodedChar: 
EncodedChar: | EncodedCode: 175 | DecodedCode: 97 | DecodedChar: a
EncodedChar:  | EncodedCode: 184 | DecodedCode: 255 | DecodedChar: 
 | EncodedCode: 13 | DecodedCode: 61 | DecodedChar: =
EncodedChar: | | EncodedCode: 124 | DecodedCode: 57 | DecodedChar: 9
EncodedChar: p | EncodedCode: 112 | DecodedCode: 238 | DecodedChar:
EncodedChar: 2 | EncodedCode: 50 | DecodedCode: 117 | DecodedChar: u
EncodedChar:  | EncodedCode: 208 | DecodedCode: 236 | DecodedChar:
ȝxX  ZC  3F}?Bk8an 9!|vT"a=9u

5. Выводы по работе.

В ходе данной работы был изучен метод шифрования данных методом линейного шифрования данных. Также были получены практические навыки использования данного метода.